Bacterial Vaginosis Is Associated with Variation in Dietary Indices

Bacterial vaginosis (BV) is a common condition of unknown etiology and has been linked to adverse reproductive and obstetric health outcomes. Prior dietary research on BV has focused on specific macro- and micronutrients, but not dietary indices. We assessed the relationship between BV and selected dietary indicators among a cohort of 1735 nonpregnant women ages 15–44 y from Birmingham, Alabama. Annual intake was assessed with the Block98 FFQ, and the glycemic index, glycemic load (GL), and Healthy Eating Index were calculated by the Block Dietary Data System. The Naturally Nutrient Rich (NNR) score was also calculated. Vaginal flora was evaluated using Nugent Gram-stain criteria. Crude OR and adjusted OR were determined by multinomial and logistic regression in cross-sectional and prospective analyses, respectively. Participants were predominantly African American (85.5%) aged 25.3 ± 6.8 y (mean ± SD). Per 10-unit increase, GL was positively (adjusted OR = 1.01, 95% CI = 1.00–1.03) and NNR was negatively (adjusted OR = 0.93, 95% CI = 0.88–0.99) associated with BV compared to normal vaginal flora. In prospective analyses, only GL was associated with BV progression (adjusted OR = 1.03, 95% CI = 1.00–1.05) and persistence (adjusted OR = 1.02, 95% CI = 1.01–1.04) after adjustment. Both GL and NNR were associated with greater BV prevalence and GL was associated with an increase in BV persistence and acquisition. These results suggest that diet composition may contribute to vaginal flora imbalances and be important for elucidating the etiology of BV.